# 10. COLUMNS AND STRUTS

## 10.1 Introduction:

In many incidents we see the members subject to compressive loads such as truss, building structures etc. A structural member subjected to axial compressive force is called a strut. A strut may be horizontal or inclined. If he member is vertical & subjected to axial compressive force is termed as column, pillar or stanchion.

## 10.2 Axially Loaded Compressive Members: Elastic Stability:

When a column or strut is subjected to some compressive force, then the compressive stress induced

$$\sigma = \frac{P}{A} \qquad \text{…..(10.1)}$$

Where P = Compressive Force

A = Cross – Sectional area of the column.

If the column is short it fails by compressive stress, but if the column is long and subjected to a compressive load it fails under bending or buckling. The load at which the column just buckles is called buckling load, critical load or crippling load and the column is said to have an elastic instability. Equilibrium of an absolutely original body may be stable, neutral and unstable.

If a long column is subjected to an axial force P (less than a certain critical load $P_{cr}$) it will compress a little and will remain straight. The column is in a condition of stable equilibrium.

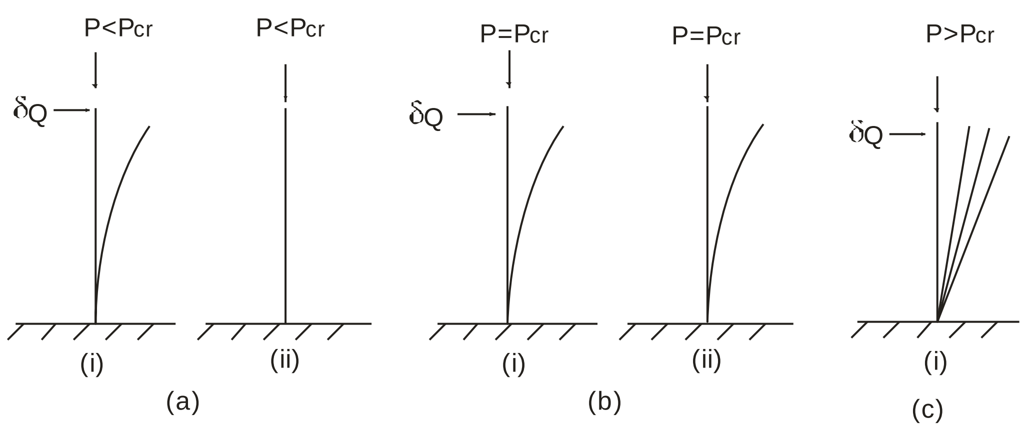


Fig. 10.1

Fig. 10.1(a) If it is deflected slightly by some transverse load $\delta$Q and then the load is removed the column become straight again and will assume the original condition. The Column is said to be in stable condition. For the case Fig. 10.1(b) the applied load is equal to the critical load. If the lateral load $\delta$Q is applied and removed after deflection, the column remains in deformed position after removal of $\delta$Q. This instantaneous condition is condition of Neutral Equilibrium. Finally if he load is increased beyond the critical load value, the straight configuration of equilibrium will become unstable and the column will continue to deform after the removal of the lateral load Fig. 10.1(c).

This instability is to be avoided in the design and hence knowledge of the critical load for the member becomes necessary.

## 10.3 Classification Of Columns:

Depending upon the ratio between the length and diameter column and slenderness ratio (ratio of length of the column to the minimum radius of gyration of the cross sectional area of the column) can be divided into three classes:

**(a) Short Column:**

Columns which have length less than 8 times their diameter or slenderness ratio is less than 32 are called short column or stocky struts. In short column buckling is negligible and hence short columns are always subjected to direct compressive stresses only.

(b) **Medium Size Column:**

Columns which have length diameter l/d ratio 8 to 30 or slenderness ratio 32 to 120 are called medium size columns. In this case buckling and direct stresses both are to be taken into account while designing.

(c) **Long Columns:**

Column which have length diameter ratio l/d more than 30 or slenderness ratio less than 12 are called long columns. In this case direct compressive stress is much less than buckling stress.

## 10.4 Euler’s Theory - Long Columns:

**Assumptions:**

1. The column is very long in proportions to its cross sectional dimensions.
2. The section of column is uniform.

3. The column is assumed to be perfectly straight.

4. Column material is homogeneous in quality, isotropic and perfectly elastic.

5. The direct stress is very small as compared to bending stress.

6. The compressive load is applied perfectly axially.

7. The self weight of column is ignorable.

8. The column will fail by buckling alone.

**Buckling Load**: When a column is loaded and the load is steadily increased a certain limiting value of the load will be reached when the column will just tend to deflect or have a lateral displacement. At this position the internal forces which tend to straighten the column are just equal to the applied load. The minimum limiting load at which column ends to have lateral displacement or tends to buckle is known as buckling load, crippling load or critical load. Buckling takes place about the axis having the least moment of inertia.

The strength to resist buckling is greatly affected by the condition of the ends. The following four cases arise:

(1) Both ends are hinged.

(2) One end is fixed and other is free.

(3) Both ends are fixed

(4) One end is fixed and other is hinged.

**10.4.1 Both Ends Hinged:**

Let us consider a column AB with ends free to rotate around frictionless pins i.e. ends are hinged carrying a critical load at B. The buckled shape is shown in Fig. 10.2 at the critical load. The figure is possible only at critical load and prior to this load column remains straight.

Let us consider a section at a distance $x$ from A and let y be its deflection from the center line. We have therefore, the bending moment at the section is

$$EI \frac{d^2y}{dx^2} = M_x = -Py$$

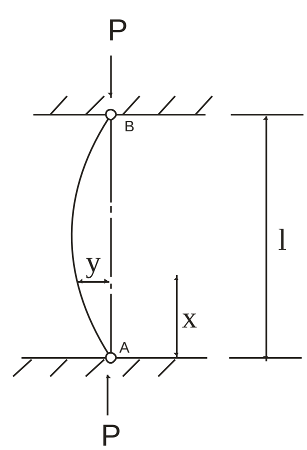


Fig.10.2

[ -ve sign has been taken as bend is concave to original line ]

$$\frac{d^2y}{dx^2} + \frac{P}{EI}y = 0$$

This is a standard differential equation, the solution is

$$y = C_1 \cos x\sqrt{\frac{P}{EI}} + C_2 \sin x\sqrt{\frac{P}{EI}} \quad \text{.....(10.2)}$$

where $C_1$ and $C_2$ are constant and can be found by applying boundary conditions.

At A deflection is zero $x = 0$ & $y = 0$ $\therefore C_1 = 0$

At B also deflection is zero $x = l$ & $y = 0$

$$\therefore C_2 \sin l\sqrt{\frac{P}{EI}} = 0$$

Either $C_2 = 0$ (i.e. there is no bend in column hence $C_2$ cannot be 0 )

Hence $\sin l\sqrt{\frac{P}{EI}} = 0$

$$\therefore l\sqrt{\frac{P}{EI}} = n\pi, \quad n = 0, 1, 2, 3..... \quad \text{.....(10.3)}$$

Taking least significant value

$$l\sqrt{\frac{P}{EI}} = \pi \qquad \therefore P = \frac{\pi^2 EI}{l^2} \quad \text{.....(10.4)}$$

Hence the least critical load, buckling load or Euler load

$$P_E = \frac{\pi^2 EI}{l^2}$$

The equation (10.2) becomes

$$y = C_2 \sin x\sqrt{\frac{P}{EI}}$$

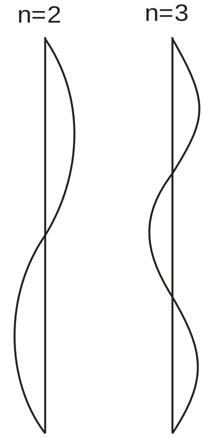

Fig.10.3

This is the characteristic or eigen function of this problem and since $\sqrt{\frac{P}{EI}} = \frac{n\pi}{L}$, n can assume any integer value , there is infinite no. of such functions. In this lineraized solution amplitude $C_2$ of buckling mode remains indeterminate. For fundamental case n = 1, the elastic curve is a half wave sine curve. Shape and modes corresponding to n = 2, 3 are shown in Fig.10.3. The higher modes have no physical significance in buckling problems since the least critical buckling load occurs at n = 1.

**10.4.2 One End Is Fixed Other End Is Free :**

Fig. 10.4 shows a column AB of length $l$ whose one end is fixed and other is free. The shape after applying buckling load P is shown in Fig. 10.4.

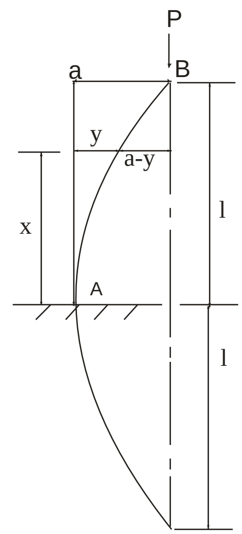

Fig. 10.4

Let 'a' be the deflection of the free end and let y be the deflection at any distance $x$ from free end A.

The bending moment at the section is given by

$$= P(a - y)$$

Hence $EI \dfrac{d^2y}{dx^2} = P(a - y)$

$$\frac{d^2y}{dx^2} + \frac{P}{EI}y = \frac{Pa}{EI}$$

the solution of the differential equation is given by

$$y = C_1 \cos x\sqrt{\frac{P}{EI}} + C_2 \sin x\sqrt{\frac{P}{EI}} + a \qquad \text{.....(10.5)}$$

$C_1$ and $C_2$ are constants of integration and are found out by applying boundary conditions.

At A deflection is zero $\quad x = 0; \quad y = 0 \quad \therefore C_1 = -a$

At A slope is zero $\quad x = 0 \quad \dfrac{dy}{dx} = 0$

$$\therefore \frac{dy}{dx} = -C_1\sqrt{\frac{P}{EI}}\sin x\sqrt{\frac{P}{EI}} + C_2\sqrt{\frac{P}{EI}}\cos x\sqrt{\frac{P}{EI}}$$

At A $\quad 0 = -a\sqrt{\dfrac{P}{EI}}\; 0 + C_2\sqrt{\dfrac{P}{EI}}\,.1$

$$C_2\sqrt{\frac{P}{EI}} = 0$$

Either P = 0 (Not applicable)

or $C_2 = 0$

in equation (10.5) $C_1 = -a$ ; $C_2 = 0$

$$y = -a \cos x\sqrt{\frac{P}{EI}} + a$$

$$y = a\left(1 - \cos x\sqrt{\frac{P}{EI}}\right)$$

at $x = l$, y = a  $$a = a\left(1 - \cos l\sqrt{\frac{P}{EI}}\right)$$

$$1 = 1 - \cos l\sqrt{\frac{P}{EI}}$$

$$\cos l\sqrt{\frac{P}{EI}} = 0$$

$$l\sqrt{\frac{P}{EI}} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

Taking the least significant value

$$l\sqrt{\frac{P}{EI}} = \frac{\pi}{2}$$

Hence the least critical load, buckling load or Euler load

$$P = P_E = \frac{\pi^2 EI}{4l^2} \qquad \text{…..(10.6)}$$

**10.4.3 Both Ends Are Fixed:**

Fig. 10.5 shows a column AB of length $l$ whose both ends are fixed. There will be a restraint moment say $M_0$ at each end. The deformed shape under the application of buckling load P has been shown in Fig. 10.5. The column is convex for a length of $\frac{l}{4}$ from either end and concave for the middle half.

Considering at any distance $x$ from A we have

$$EI\frac{d^2y}{dx^2} = M_O - P_y$$

Where $M_O$ is the fixed moment at the ends as in the case of a fixed beam

$$\frac{d^2y}{dx^2} + \frac{P}{EI}y = \frac{PM_O}{EI\ P}$$

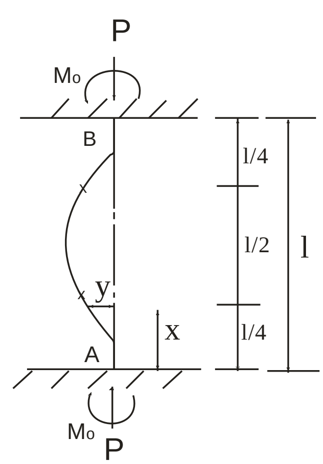

Fig. 10.5

The solution of the above differential equation is given by

$$y = C_1 \text{Cos}\, x \sqrt{\frac{P}{EI}} + C_2 \sin x \sqrt{\frac{P}{EI}} + \frac{M_O}{P} \qquad \text{.....(10.7)}$$

$$\frac{dy}{dx} = -C_1 \sqrt{\frac{P}{EI}}\, \text{Sin}\, x \sqrt{\frac{P}{EI}} + C_2 \sqrt{\frac{P}{EI}}\, \text{Cos}\, x \sqrt{\frac{P}{EI}} \qquad \text{.....(10.8)}$$

Where $C_1$ and $C_2$ are constants and found by applying boundary conditions

At A Deflection is zero $\quad x = 0; \quad y = 0 \quad \therefore\ C_1 = -\frac{M_O}{P}$

At A slope is zero $\quad x = 0 \quad \frac{dy}{dx} = 0 \quad \therefore C_2 = 0\ (P \neq 0)$

In equation (10.7) substituting $\quad C_1 = -\frac{M}{P}; \quad C_2 = 0$

$$y = C_1 \text{Cos}\, x\sqrt{\frac{P}{EI}} - C_1 = C_1 \left(\text{Cos}\, x\sqrt{\frac{P}{EI}} - 1\right)$$

at B deflection is zero $x = l$; $y = 0$; $0 = C_1 \left(\text{Cos}\, l\sqrt{\frac{P}{EI}} - 1\right)$

either $C_1 = 0$ in which case column is not bent at all

or $\quad \text{Cos}\, l \sqrt{\frac{P}{EI}} = 1$

$$l\sqrt{\frac{P}{EI}} = 0, 2\pi, 4\pi \qquad \text{.....(10.9)}$$

also at B slope is zero $x = 1$, $\frac{dy}{dx} = 0$

from equation (10.8)

$$0 = -C_1 \sqrt{\frac{P}{EI}}\, \text{Sin}\, l \sqrt{\frac{P}{EI}}$$

$$\text{Sin}\, l \sqrt{\frac{P}{EI}} = 0$$

$$l \sqrt{\frac{P}{EI}} = 0, \pi, 2\pi, 3\pi \ldots \qquad \text{.....(10.10)}$$

The minimum significant value of $l\sqrt{\frac{P}{EI}}$ consistent both with (10.9) & (10.10) is $2\pi$.

$$\therefore\ l\sqrt{\frac{P}{EI}} = 2\pi$$

Hence the least critical load, buckling load or Euler load

$$P = P_E = \frac{4\pi^2 EI}{l^2} \qquad \text{.....(10.11)}$$

**10.4.4 One End Is Fixed, The Other Is Hinged:**

Fig. 10.6 shows the deformed shape of a column which is fixed at one end A and hinged at B carrying a buckling load P. Since A is fixed a bending moment $M_o$ is induced at A and this involves the presence of a force R at right angle to AB (i.e. horizontal) to maintain equilibrium, without which no BM can occur at A. Let us consider any section at a distance *x* from A the bending moment at section is given by

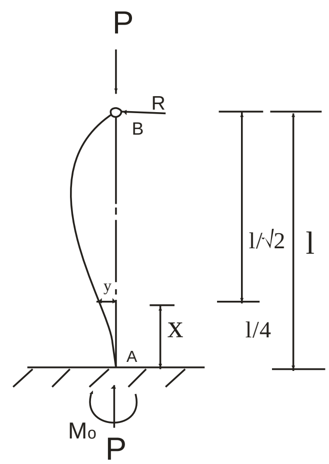


Fig. 10.6

$$EI\frac{d^2y}{dx^2} = -P_y + R(l-x)$$

$$\frac{d^2y}{dx^2} + \frac{P}{EI}y = \frac{R}{EI}(l-x)$$

The solution of the above differential equation is given by

$$y = C_1 \text{Cos}\, x\sqrt{\frac{P}{EI}} + C_2 \text{Sin}\, x\sqrt{\frac{P}{EI}} + \frac{R(l-x)}{P} \qquad \text{.....(10.12)}$$

$$\frac{dy}{dx} = -C_1\sqrt{\frac{P}{EI}}\ \text{Sin}\, x\sqrt{\frac{P}{EI}} + C_2\sqrt{\frac{P}{EI}}\ \text{Cos}\, x\sqrt{\frac{P}{EI}} - \frac{R}{P} \qquad \text{.....(10.13)}$$

At A deflection is zero $\quad x = 0 \quad y = 0 \quad \therefore C_1 = -\frac{Rl}{P}$

At A slope is zero $\quad x = 0 \quad \frac{dy}{dx} = 0 \quad \therefore 0 = C_2\sqrt{\frac{P}{EI}} - \frac{R}{P}$

$$\therefore \quad C_2 = \frac{R}{P}\sqrt{\frac{EI}{P}}$$

In equation (10.12)

$$y = -\frac{Rl}{P}\operatorname{Cos} x\sqrt{\frac{P}{EI}} + \frac{R}{P}\sqrt{\frac{EI}{P}}\operatorname{Sin} x\sqrt{\frac{P}{EI}} + \frac{R}{P}(l - x)$$

at B deflection is zero $\quad x = l \quad y = 0$

$$0 = -\frac{Rl}{P}\operatorname{Cos} l\sqrt{\frac{P}{EI}} + \frac{R}{P}\sqrt{\frac{EI}{P}}\operatorname{Sin} l\sqrt{\frac{P}{EI}} + 0$$

$$\operatorname{Tan} l\sqrt{\frac{P}{EI}} = l\sqrt{\frac{P}{EI}}$$ {such that tangent is equal to itself}

The smallest root of above equation

$$l\sqrt{\frac{P}{EI}} = 4.49 \text{ radian} \quad \therefore \quad \frac{l^2 P}{EI} = 2\,0 = 2\pi^2 \text{ (approximately)}$$

Hence the least critical load, buckling load or Euler load

$$P = P_E = \frac{2\pi^2 EI}{l^2} \qquad \text{.....(10.14)}$$